Object Oriented Programming
(Assignment Solutions)

Question 1 :

//Only overloaded Operator function is given, rest is already covered in Lecture.

```
Complex operator - (Complex &obj) {
    int resReal = this->real - obj.real;
    int resImg = this->img - obj.img;
    Complex res(resReal, resImg);
    return res;
}
```

Question 2 :

```
class BankAccount {
  private:
    int accountNumber;
    double balance;

  public:
    BankAccount(int accNum, double bal) {
        accountNumber = accNum;
        balance = bal;
    }

    void deposit(double amount) {
        balance += amount;
    }

    void withdraw(double amount) {
        if(amount <= balance) {
            balance -= amount;
        } else {
            cout << "No sufficient balance for this withdrawal.\n";
        }

    }

    double getBalance() {
```

```
            return balance;
        }
};


int main() {
   BankAccount myAccount(123456, 500.0);
   myAccount.deposit(150.0);
   myAccount.withdraw(100.0);
   cout << "Current Balance: Rs." << myAccount.getBalance() << endl;
   return 0;
}
```

Question 3 :

```
class Person {
protected:
        string name;
        int age;

public:
        Person(string n, int a) {
           name = n;
           age = a;
        }
};

class Student : public Person {
private:
        string studentID;

public:
        Student(string n, int a, string id) : Person(n, a) {
            studentID = id;
        }

        void displayStudentInfo() {
            cout << "name : " << this->name << endl;
            cout << "age : " << this->age << endl;
            cout << "studentId : " << this->studentID << endl;
        }
};
```

```cpp
int main() {
    Student student("Alice", 20, "S12345");
    student.displayStudentInfo();
    return 0;
}
```

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